

EXCITATION-DEPENDENT IDENTIFIABILITY OF LATENT HIGHER-ORDER STRUCTURE FROM EDGE FLOWS

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ABSTRACT

Which triangles of a network carry higher-order interactions? Edge flows reveal the latent filled-triangle set $\mathcal{S}$ only through their curl, and we show that *what* is identifiable is governed by the excitation covariance $\Gamma_{\mathcal{S}}$ of the triangle signals, through three regimes. For positive-diagonal $\Gamma_{\mathcal{S}}$ (known or unknown), the weighted support is identifiable at *any* rank deficiency of $\mathbf{B}_2$, because the lifted signatures $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}$ are always linearly independent. For unrestricted PSD $\Gamma_{\mathcal{S}}$, the population curl covariance determines $\mathbf{M} = \Sigma_z - \sigma_n^2 \mathbf{I}$ exactly but pins the support only up to $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$ (the achievable set is $\mathcal{C}_{\mathcal{S}} = \{\mathbf{M} \succeq 0 : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$); a strictly positive-diagonal but correlated $\Gamma_{\mathcal{S}}$ already defeats identification (a rank-one witness on K_4). At the projector excitation $\Gamma_{\mathcal{S}} = \sigma_c^2 (\mathbf{B}_{2,\mathcal{S}}^\top \mathbf{B}_{2,\mathcal{S}})^+$ the covariance collapses to $\sigma_n^2 \mathbf{I} + \sigma_c^2 \mathbf{P}_{\text{im}}$ exactly, so equal-image supports are indistinguishable at every SNR and sample size (noise level known throughout). This projector is exactly the Hodge-smoothness prior used in prior topology learning [1]. In the diagonal regime we give a lifted-covariance non-negative least-squares estimator that is consistent with a fully explicit $O(1/N)$ failure bound, validated on K_4 – K_8 against subspace and greedy baselines, including an excitation interpolation under which equal-image distinguishability provably vanishes. On real data, the same curl statistic *localizes* genuine unplanted rotational structure — tropical cyclones in ERA5 winds — against a same-field vorticity reference and an external IBTrACS archive. Code, data, and a 58-test guardrail suite (the quoted constants checked against Monte-Carlo) are released.

Index Terms— Topological signal processing, simplicial complexes, identifiability, Hodge decomposition, covariance estimation, edge flows.

1. INTRODUCTION

Edge flows — currency exchange, traffic, ranking comparisons — carry higher-order structure on the triangles of a simplicial complex [2, 3, 4]. Topological signal processing builds filters and detectors on the Hodge Laplacian [5, 6, 7, 8, 9], and a recurring primitive is that the set $\mathcal{S}$ of “filled” triangles is unknown. One line *estimates* $\mathcal{S}$ algorithmically (MAP under a Hodge-smoothness prior [1], greedy learning [10], sparse cell-complex augmentation [11]); a second-order line models the edge covariance (PCA/total-variation on the sample covariance [2, §VII], ML fitting of a Hodge-structured simplicial Gaussian model [12]); a third *detects* signals in a *known* Hodge subspace [13]. Each fixes a model and gives an algorithm; to our knowledge none characterizes, as a function of the excitation class, *when* the filled-triangle structure is identifiable from the edge

covariance — the gap this paper fills: *what part of the latent structure is identifiable, and when?*

The answer is not a property of the geometry alone: it depends on the *excitation*. We model triangle signals as $\mathbf{y}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \Gamma_{\mathcal{S}})$ with PSD excitation covariance $\Gamma_{\mathcal{S}}$ and show that the identifiable object moves with the regime of $\Gamma_{\mathcal{S}}$ (Theorem 1): from the full weighted support (positive diagonal, any rank deficiency), to only the achievable-covariance set $\mathcal{C}_{\mathcal{S}}$ — so the support merely up to $\{\mathcal{S} : \text{im } \mathbf{M} \subseteq \text{im } \mathbf{U}_{\mathcal{S}}\}$ — under unrestricted PSD excitation, to provable indistinguishability of equal-image supports at the projector excitation. The last case is not a corner curiosity: the Hodge-smoothness prior of [1], $\mathbf{y}_{\mathcal{S}} \sim \mathcal{N}(\mathbf{0}, \mathbf{L}_2^+)$, $\mathbf{L}_2 = \mathbf{B}_2^\top \mathbf{B}_2$, is this projector excitation — equal-image supports then have identical marginalized covariance likelihood, so any choice between them comes from the prior, not the data. We thus correct a folklore “rank obstruction” and delimit what covariance- and image-level methods [2, 12, 13] can resolve. Our tools are classical — detection theory and Fano arguments [14, 15], in the lineage of support-recovery and sparse-covariance limits [16, 17, 18, 19] — applied through a reduction (curl annihilation, after HodgeRank’s curl-as-inconsistency reading [20]) that makes them bite. Contributions:

1. *Theory*: the three excitation regimes (Thm. 1), with global analytic separation constants for the isotropic regime — equal-image binary supports are separated by at least 9 (attained by adding a tetrahedron’s fourth face) and equal-cardinality supports by 16 (swaps), proved for *every* clique complex via a Johnson-graph eigenvalue argument — and Chernoff exponents, for fixed graph and support as $\rho_2 \rightarrow 0$, whose worst case equals one isolated-triangle detection (Thm. 2).
2. *Achievability*: a lifted-covariance NNLS estimator for the diagonal regime, consistent with a fully explicit $O(1/N)$ failure bound in which the constants are derived and numerically spot-checked (Thm. 3); Monte-Carlo validation on K_4 – K_8 with random supports against oracle-aided subspace and greedy baselines, and an excitation-interpolation experiment in which equal-image distinguishability collapses as predicted (Fig. 1).
3. *Real data, carefully positioned*: curl-based *vortex localization* — not topology recovery — of unplanted tropical-cyclone circulation in ERA5 winds, validated against a same-field vorticity reference and the independent IBTrACS archive, with a classical baseline and moving-block bootstrap confidence intervals (Fig. 2).

2. MODEL AND CURL REDUCTION

Fix a graph $(\mathcal{V}, \mathcal{E})$ with node–edge incidence $\mathbf{B}_1$ and, for a candidate triangle set $\mathcal{T}$ (3-cliques), edge–triangle incidence $\mathbf{B}_2 \in \mathbb{R}^{|\mathcal{E}| \times p}$;

$B_1 B_2 = \mathbf{0}$, and each column $\mathbf{b}_\tau$ has three ± 1 entries, so $\mathbf{G} = B_2^\top B_2$ has $G_{\tau\tau} = 3$ and $G_{\sigma\tau} \in \{0, \pm 1\}$ (nonzero iff the triangles share an edge). We observe N i.i.d. snapshots

$$\mathbf{f}_t = B_1^\top \mathbf{a}_t + B_{2,S} \mathbf{y}_t + \mathbf{h}_t + \mathbf{n}_t, \quad t = 1, \dots, N, \quad (1)$$

with gradient nuisance $\mathbf{a}_t \sim \mathcal{N}(\mathbf{0}, \sigma_g^2 \mathbf{I})$, harmonic nuisance $\mathbf{h}_t$, noise $\mathbf{n}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{I})$, and triangle excitation

$$\mathbf{y}_t \sim \mathcal{N}(\mathbf{0}, \Gamma_S), \quad \Gamma_S \succeq \mathbf{0}, \quad (2)$$

where $\Gamma_S \succeq \mathbf{0}$ is the excitation covariance; making it explicit is the point of this paper (prior generative models are special cases of it, §1). We take the harmonic nuisance to be *candidate-orthogonal*, $\mathbf{h}_t \in \ker B_2^\top$ (orthogonal to the curl space of the *full* candidate complex); the gradient part is annihilated unconditionally since $B_2^\top B_1^\top = \mathbf{0}$. Then curl projection removes both nuisances exactly [20]: in an orthonormal basis $\mathbf{Q}$ of $\text{im } B_2$ ($r = \text{rank } B_2$, B_2 the full candidate incidence), the sufficient statistic $\mathbf{z}_t = \mathbf{Q}^\top \mathbf{f}_t$ is Gaussian with

$$\Sigma_z(S, \Gamma_S) = \sigma_n^2 \mathbf{I}_r + U_S \Gamma_S U_S^\top, \quad U = \mathbf{Q}^\top B_2, \quad (3)$$

with $\|\mathbf{u}_\tau\|^2 = 3$ and $\mathbf{u}_\sigma^\top \mathbf{u}_\tau = G_{\sigma\tau}$. Identifiability of S from (3) is the object of study; for the isolated-triangle scalar test this reduces to the classical two-variance problem with curl-SNR $\rho = 3\sigma_c^2/\sigma_n^2$ [14] (supplement §S1–S3 develops that line: detection thresholds, Fano converses, decorrelated detectors, plug-in estimation).

Definition 1 (Observational equivalence and identifiability). *All statements here are about the population Σ_z (finite- N estimation is §4). Two pairs (S, Γ_S) and $(S', \Gamma_{S'})$ are observationally equivalent if $\Sigma_z(S, \Gamma_S) = \Sigma_z(S', \Gamma_{S'})$; a functional of (S, Γ_S) is identifiable on an excitation class $\mathcal{G}$ if it is constant on each equivalence class with excitations in $\mathcal{G}$. The realized range is $\mathcal{R}(S, \Gamma_S) = \text{im}(\Sigma_z - \sigma_n^2 \mathbf{I}) = \text{im}(U_S \Gamma_S^{1/2}) \subseteq \text{im } U_S$, always identifiable since it is a function of Σ_z alone. If $\Gamma_S \succ \mathbf{0}$ then $\mathcal{R} = \text{im } U_S$; when $B_{2,S}$ has full column rank (so U_S is injective) the converse holds as well.*

3. EXCITATION-DEPENDENT IDENTIFIABILITY

Lemma 1 (Lifted spark). *For any candidate set of distinct 3-cliques of a simple graph, the matrices $\{\mathbf{b}_\tau \mathbf{b}_\tau^\top\}_{\tau \in \mathcal{T}}$ are linearly independent.*

Proof. A pair of distinct edges lies in at most one common triangle, so for $e \neq e'$ both in τ , the (e, e') entry of $\sum_\sigma w_\sigma \mathbf{b}_\sigma \mathbf{b}_\sigma^\top$ equals $\pm w_\tau$: every coefficient is read off a distinct matrix entry. $\square$

Theorem 1 (Three excitation regimes). *Let σ_n^2 be known and write $\mathbf{M} = \mathbf{M}(S, \Gamma) = U_S \Gamma U_S^\top = \Sigma_z - \sigma_n^2 \mathbf{I}$ in (3). The regimes below are overlapping constraints on Γ_S , not a partition (e.g. $\sigma_c^2 \mathbf{I}$ is both diagonal (a) and a special full-rank PSD (b)). (a) Positive diagonal. If $\Gamma_S = \text{diag}(\gamma_\tau)_{\tau \in S}$ with $\gamma_\tau > 0$ (known or unknown), then $\mathbf{M} = \sum_{\tau \in S} \gamma_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top$ and, by Lemma 1, the map $(S, \Gamma_S) \mapsto \Sigma_z$ is injective: the weighted support is identifiable at any rank deficiency of B_2 , including all $\binom{n}{3}$ triangles of K_n . (b) Arbitrary PSD. The achievable set $\mathcal{C}_S = \{U_S \Gamma U_S^\top : \Gamma \succeq \mathbf{0}\} = \{\mathbf{M} \succeq \mathbf{0} : \text{im } \mathbf{M} \subseteq \text{im } U_S\}$, so the population covariance determines $\mathbf{M}$ in full but pins the support only to $\{S : \text{im } \mathbf{M} \subseteq \text{im } U_S\}$: the realized range $\mathcal{R} = \text{im } \mathbf{M} \subseteq \text{im } U_S$ is always readable, yet the candidate image $\text{im } B_{2,S}$ is not (a larger-image S' with a suitable Γ gives the same*

$\mathbf{M}$; $\Gamma_S \succ \mathbf{0}$ does recover $\mathcal{R} = \text{im } U_S$). The obstruction is correlation, not zero variance: on K_4 (all four faces) with $\mathbf{c} \in \ker U_S$ and $\mathbf{v} = \mathbf{e}_1 - \frac{1}{4}\mathbf{c} = (.75, .25, -.25, .25)^\top$, the strictly-positive-diagonal $\Gamma' = \mathbf{v}\mathbf{v}^\top$ gives $U_S \Gamma' U_S^\top = \mathbf{u}_1 \mathbf{u}_1^\top$ — identical to the singleton $\{\tau_1\}$ though the images are 3- vs. 1-dimensional. Regime (a) additionally requires Γ diagonal. (c) Projector excitation. For $\Gamma_S = \sigma_c^2 (B_{2,S}^\top B_{2,S})^+$, $\mathbf{M} = \sigma_c^2 P_{\text{im } U_S}$ exactly; supports with equal curl image induce identical snapshot distributions and are indistinguishable at every SNR and every N .

Proof sketch. (a) Read the weights off the lifted representation (spark). (b) $\subseteq$ is immediate; $\supseteq$: given PSD $\mathbf{M}$ with $\text{im } \mathbf{M} \subseteq \text{im } U_S$, take $\Gamma = U_S^\top \mathbf{M} (U_S^\top)^\top \succeq \mathbf{0}$. (c) $\mathbf{A}(\mathbf{A}^\top \mathbf{A})^+ \mathbf{A}^\top = P_{\text{im } \mathbf{A}}$ (SVD). On K_n , $\dim \text{im } B_2 = \binom{n-1}{2}$ (a $3/n$ DoF ratio), and equal-image supports exist whenever a tetrahedron is hosted ($\mathbf{b}_{012} - \mathbf{b}_{013} + \mathbf{b}_{023} - \mathbf{b}_{123} = \mathbf{0}$). Full proofs and numerical certificates: supplement §S4, tests/test_excitation.py. $\square$

Remark 1 (Scope). (i) *Deterministic unknown signals are the rank-one degenerate boundary of regime (b): the realized range is a single direction, so not even the image is seen.* (ii) *On K_n with σ_n unknown, regime (a) retains exactly one ambiguous direction, since $\sum_{\tau \in \mathcal{T}} \mathbf{u}_\tau \mathbf{u}_\tau^\top = n \mathbf{I}_r$: a uniform weight shift trades off against the noise floor.* (iii) *Regime (b) delimits image-level methods — e.g. matched subspace detectors [13] — whose population statistics see only the realized range $\text{im } \mathbf{M} \subseteq \text{im } B_{2,S}$: without excitation structure, no method can do better.*

Theorem 2 (The price in the isotropic class). *Let $\Gamma_S = \sigma_c^2 \mathbf{I}$, $\rho_2 = \sigma_c^2/\sigma_n^2$, and write the dimensionless signature Gram $\mathbf{D}_S = \sum_{\tau \in S} \mathbf{u}_\tau \mathbf{u}_\tau^\top$ (so $\mathbf{M}_S = \sigma_c^2 \mathbf{D}_S$). For any clique complex: distinct binary supports with equal curl image satisfy $\|\mathbf{D}_S - \mathbf{D}_{S'}\|_F^2 \geq 9$, with equality exactly for single-face differences (e.g. $S' = S \cup \{\text{fourth face}\}$), and ≥ 16 under equal cardinality (tetrahedral swaps). For fixed graph and support pair, as $\rho_2 \rightarrow 0$ the optimal-test Chernoff exponents are $C_G = \rho_2^2 \|\Delta \mathbf{D}\|_F^2 / 16 (1 + o(1))$, i.e. $\frac{9}{16} \rho_2^2 (1 + o(1))$ (subset) and $\rho_2^2 (1 + o(1))$ (swap) — and $\frac{9}{16} \rho_2^2$ equals the isolated-triangle detection exponent $C(\rho) = \rho^2 / 16$ at $\rho = 3\rho_2$: the hardest equal-image decision costs one isolated-triangle detection. Hence in this scope (fixed pair, isotropic excitation, leading low-SNR exponent) rank deficiency is free at the exponent level, and only the Fano log-multiplicity of the $4 \binom{n}{4}$ tetrad hypotheses ($\asymp \log n$ on K_n) reflects the geometry.*

Proof sketch. Separations: $\|\sum_\tau c_\tau \mathbf{b}_\tau \mathbf{b}_\tau^\top\|_F^2 = \mathbf{c}^\top (9\mathbf{I} + \mathbf{A}) \mathbf{c}$ for ternary $\mathbf{c}$, where $\mathbf{A}$ is the share-an-edge adjacency — an induced subgraph of the Johnson graph $J(n, 3)$, so $\lambda_{\min}(\mathbf{A}) \geq -3$ by interlacing and $\text{sep} \geq 6\|\mathbf{c}\|^2$; enumerate $\|\mathbf{c}\|^2 \in \{1, 2\}$. Exponents: Bhattacharyya expansion gives $C_G = \rho_2^2 \|\Delta \mathbf{D}\|_F^2 / 16 (1 + o(1)) = \|\Delta \mathbf{M}\|_F^2 / (16\sigma_n^4) (1 + o(1))$ (the two forms coincide). Numerically checked: swap C_G/ρ_2^2 within 6×10^{-4} of 1 and subset matching the single-triangle exponent to 3×10^{-5} relative (at $\rho_2 = 10^{-4}, 10^{-5}$); separations exhaustive over all 2^{10} supports of K_5 . Proofs: supplement §S4. $\square$

Proposition 1 (Edge-sharing marginal laws). *Assume B_2 full column rank and $\Gamma_S = \sigma_c^2 \mathbf{I}$. The GLS/BBLUE inversion $\hat{\mathbf{y}}_t = \mathbf{G}^{-1} B_2^\top \mathbf{f}_t$ [21] satisfies, exactly, $\hat{\mathbf{y}}_t = \tilde{\mathbf{y}}_t + \mathbf{e}_t$ with $\mathbf{e}_t \sim \mathcal{N}(\mathbf{0}, \sigma_n^2 \mathbf{G}^{-1})$: each coordinate is a two-variance test with effective SNR $\rho_{\tau}^{\text{eff}} = \sigma_c^2 / (\sigma_n^2 (\mathbf{G}^{-1})_{\tau\tau})$ — the classical variance-inflation factor as the price of edge-sharing — and exact recovery admits a correlation-robust union-bound guarantee. The noise coordinates stay correlated ($\sigma_n^2 \mathbf{G}^{-1}$ is not diagonal), so the joint law does not*

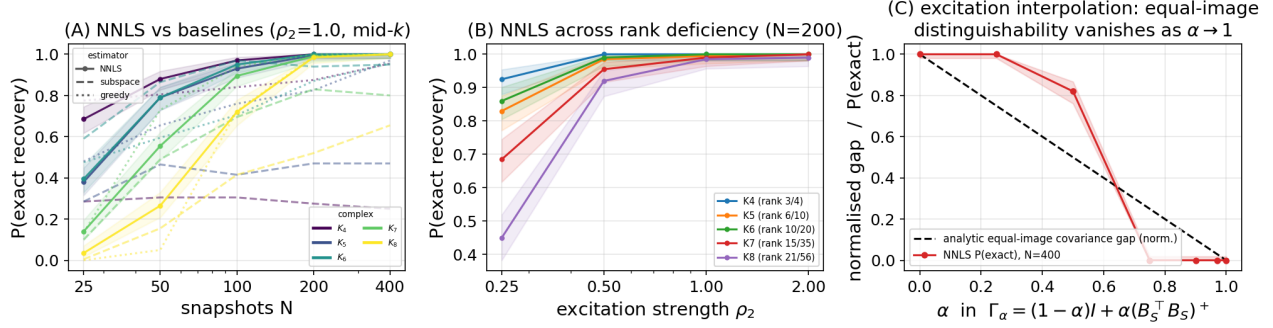


Fig. 1. Second-order achievability. (A) Exact recovery vs. N on K_4 – K_8 (random supports, $\rho_2 = 1$): NNLS (solid, 95% Wilson bands) vs. oracle-aided subspace (dashed) and greedy (dotted) baselines. (B) NNLS vs. ρ_2 at $N = 200$ across rank deficiency (legend: rank B_2/p). (C) Excitation interpolation on the K_5 tetrad: the analytic equal-image gap (normalized) closes only at $\alpha = 1$ (Thm. 1(c)); thresholded NNLS recovery collapses earlier (zero from $\alpha = 0.75$).

factorize; benchmarks, the naive-detector failure mode, and proofs are in supplement §S1, S3.

Proposition 2 (Detection and joint-recovery scalings). *In the isotropic class, deciding one isolated triangle is the two-variance Gaussian test with variances $3\sigma_n^2$ vs. $9\sigma_c^2 + 3\sigma_n^2$ [14]: attaining error $\leq \delta$ at the exponent level requires the curl-SNR scale $\rho \geq \rho^*(N) = \Theta(\sqrt{\log(1/\delta)/N})$ ($C(\rho) = \rho^2/16 + o(\rho^2)$ [15]). Recovering a uniformly random size- k support among p edge-disjoint candidates with error $\leq \epsilon$ requires $N \geq [(1 - \epsilon) \log \binom{p}{k} - \log 2]/(k \text{KL}_1)$, $\text{KL}_1 = (\rho - \log(1 + \rho))/2$ (Fano): joint recovery pays a $\log \binom{p}{k}$ factor. Signal-agnostic and sub-Gaussian variants, the exact edge-disjoint product law, partial edge observation (q^{3k} laws), and plug-in calibration are developed in supplement §S1–S3; these scalings are classical [16] — what is problem-specific is the reduction that produces ρ and the constants (3, 9).*

4. ACHIEVABILITY: LIFTED-COVARIANCE NNLS

In class (a) the theory says the weighted support is identifiable; the matching estimator is non-negative least squares on the lifted dictionary $\mathbf{A} = [\text{vec}(\mathbf{u}_\tau \mathbf{u}_\tau^\top)]_\tau$:

$$\hat{\mathbf{w}} = \arg \min_{\mathbf{w} \geq 0} \left\| \hat{\Sigma}_z - \sigma_n^2 \mathbf{I} - \sum_\tau w_\tau \mathbf{u}_\tau \mathbf{u}_\tau^\top \right\|_F^2, \quad (4)$$

with $\hat{\Sigma}_z = \frac{1}{N} \sum_t \mathbf{z}_t \mathbf{z}_t^\top$, $\hat{S} = \{\tau : \hat{w}_\tau > w_{\min}/2\}$, and $w_{\min} = \min_{\tau \in S} w_\tau$.

Theorem 3 (Consistency and explicit failure bound). $\hat{\mathbf{w}} \rightarrow \mathbf{w}$ almost surely, and for every N ,

$$\Pr[\hat{S} \neq S] \leq \frac{16[(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]}{N \sigma_{\min}^2(\mathbf{A}) w_{\min}^2}. \quad (5)$$

Proof sketch. Three derived steps, each numerically checked: (i) cone-constrained least squares obeys the deterministic bound $\|\hat{\mathbf{w}} - \mathbf{w}\|_2 \leq 2\|\hat{\Sigma}_z - \Sigma_z\|_F / \sigma_{\min}(\mathbf{A})$ — optimality of $\hat{\mathbf{w}}$, feasibility of $\mathbf{w}$, Cauchy–Schwarz — with $\sigma_{\min}(\mathbf{A}) > 0$ by Lemma 1; (ii) for Gaussian snapshots $\mathbb{E}\|\hat{\Sigma}_z - \Sigma_z\|_F^2 = [(\text{tr } \Sigma_z)^2 + \|\Sigma_z\|_F^2]/N$ exactly (Wishart second moments); (iii) Markov at threshold $w_{\min}/2$. The bound is conservative (Markov) — its value is the explicit $O(1/N)$ rate with computable constants; empirically the transition occurs roughly an order of magnitude earlier (Fig. 1). Proof: supplement §S4. $\square$

Fig. 1 validates the estimator across K_4 – K_8 with per-trial random supports at several sparsities k , a full (N, ρ_2) grid with $N \in \{25, \dots, 400\}$ and $\rho_2 \in \{0.25, \dots, 2\}$, 200 Monte-Carlo trials per cell, and 95% Wilson intervals (run_second_order.py; every cell, including the derived bound (5) evaluated at that cell, is in the released JSON). Protocol: the headline threshold is the theorem’s $w_{\min}/2$ (planted weights known); an oracle-free largest-gap rule is logged alongside but is worse (grid-mean 0.63 vs. 0.69, ahead in 18/260 cells). The full edge-domain pipeline is used; the greedy comparator is a generic covariance-atom heuristic, and both baselines get oracle side information (subspace: true k and $\dim \mathbf{B}_{2,S}$; greedy: true σ_n).

(A) Among these three implementations (not tuned literature methods), NNLS is the only one with exact recovery in every cell — incl. maximal rank deficiency (rank $B_2/p = 21/56$ on K_8) — reaching 1 at mid- k by $N=400$, $\rho_2=1$ (0.99 at the densest K_8 cell, $k=28$). The oracle- k subspace baseline can lead at small N but collapses on dependent candidates (0.00 at K_8 , $k=28$): its population scores tie there (Remark 1(iii)), and its finite- N tie-breaking rides on sample eigen-anisotropy, a side channel the projector excitation removes entirely. Greedy stalls at sparse supports (0.43 on K_5 at $k=2$, $N=400$). (B) Recovery improves monotonically in ρ_2 uniformly across rank deficiency — the empirical face of Theorem 1(a). The bound (5) upper-bounds every cell; being a Markov bound it is conservative by one to three orders of magnitude (the empirical failure decays exponentially, the bound as $1/N$). It is non-vacuous (clipped value < 1) at only 7 of the 260 grid cells — the small- $w_{\min}$, small- N corner, where it falls as low as 0.42 (and to ~ 0.75 at dedicated test cells) with zero observed failures; elsewhere the $O(1/N)$ rate is the guarantee. (C) Under $\Gamma_\alpha = \rho_2[(1 - \alpha)\mathbf{I} + \alpha(\mathbf{B}_{2,S}^\top \mathbf{B}_{2,S})^+]$ on the K_5 tetrad, with the same $w_{\min}/2$ rule (the diagonal model is deliberately misspecified for $\alpha > 0$ — that is the point), the analytic equal-image gap shrinks continuously and the empirical NNLS recovery of the specific S collapses before the gap reaches zero — thresholded estimation dies earlier than distinguishability — with the threshold-free endpoint at $\alpha = 1$ on Theorem 1(c): the $m=4$ equal-image supports of the planted cardinality have identical distributions, so no estimator beats chance $1/m$; the plotted 0 is specific- S recovery, itself $\leq 1/m$. Identifiability is a property of the excitation, not the geometry alone. (A matrix-free form scaling to $p \sim 10^4$ on GPU and a non-oracle BIC threshold are in supplement §S6.)

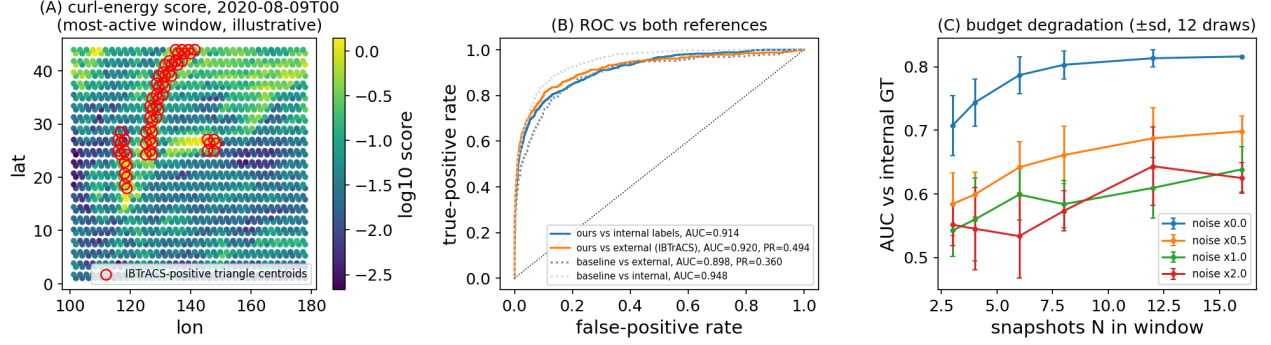


Fig. 2. Curl-based vortex localization on real data. (A) Centered curl-energy scores (vorticity scale) for the single most-active 4-day window (argmax IBTrACS-positive triangles; illustrative only — all metrics pool every window); red circles: *IBTrACS-positive triangle centroids* (not raw fix positions). (B) ROC pooled over all 15 windows: same-field vorticity reference, external IBTrACS labels (with PR-AUC), and the classical coarse-vorticity baseline. (C) Budget degradation (mean $\pm$ sd over 12 subsample draws; uniform centered statistic, $N \geq 3$).

5. REAL DATA: CURL-BASED VORTEX LOCALIZATION

We position the real-data study precisely: it is *vortex localization*, not filled-triangle topology recovery — the mesh below is simply connected, hence full-column-rank with no equal-image confusers, and real weather carries no latent boolean support. What it demonstrates is that the paper’s sufficient statistic localizes *genuine, unplanted* rotational structure on a real vector field. Construction: ERA5 10m winds [22] over the Western North Pacific (Aug–Sep 2020, $0.7^\circ/6$ -hourly) are subsampled to a 2.1° right-triangulated mesh (836 nodes, 2389 edges, 1554 triangles; $\#\text{triangles} = |\mathcal{E}| - |\mathcal{V}| + 1$ by Euler, so $\mathbf{B}_2$ is full column rank); edge flows are trapezoidal line integrals of the wind in a local equirectangular metric. By Stokes’ theorem a triangle’s curl is its circulation, so the temporally centered curl-energy score (area²-normalized to the mean-vorticity scale; no oracle parameters) ranks cyclone-bearing triangles within 4-day windows of 16 snapshots, pooled over the season’s 15 windows for evaluation. Storms translate several triangle-widths per 4-day window; centering removes the window-mean flow, so the statistic responds to the time-varying circulation a moving vortex deposits. Validation (Fig. 2B) against a *same-field consistency reference* (finite-difference vorticity on the $3\times$ -finer source grid) gives AUC 0.914 (moving-block bootstrap 95% CI [0.894, 0.936]), and against the *external, separately curated* IBTrACS archive [23] (raw: 726 fixes over 13 storms; entering the metric: 414 IBTrACS-positive of 23,310 triangle-windows, prevalence 1.8%) AUC 0.920 [0.879, 0.951], PR-AUC 0.494 [0.273, 0.666], and $P@k = 0.483$. Against a classical pointwise-vorticity baseline at the same information budget (mesh-node winds), the edge-flow statistic is better on *all three* independent external metrics (baseline: AUC 0.898, PR-AUC 0.360, $P@k = 0.389$) and slightly lower on the internal reference (0.914 vs. 0.948), which shares the baseline’s functional. The decorrelated (GLS) variant of Prop. 1 ranks worse here ($\mathbf{G}^{-1}$ amplifies noise on a near-regular mesh) and is reported in the repository. FX consistency checks and planted road-network studies from earlier revisions are retained in the repository and supplement, not claimed as recovery evidence.

Limitations. The i.i.d.-Gaussian snapshot model is an idealization of windowed weather; evaluation is therefore threshold-free ranking rather than support recovery, moving-block *and* storm-cluster bootstrap intervals (which agree) quantify pooling uncertainty and are *exploratory* (one 2020 season), and the internal reference is a

same-field consistency check — IBTrACS is the external archive. The AUC is stable across localization radius, wind cutoff, window length, and vorticity threshold (supplement). Nothing here claims a new cyclone detector; the experiment grounds the paper’s sufficient statistic on genuinely unplanted structure.

6. CONCLUSION

Identifiability of latent higher-order structure from edge flows is excitation-dependent: fully identifiable under diagonal excitation at any rank deficiency; under unrestricted PSD excitation the population covariance pins down only the achievable-covariance set $\mathcal{C}_S$, so the support is determined merely up to $\{S : \text{im } M \subseteq \text{im } U_S\}$ (not even the candidate image in general); and provably ambiguous at the projector excitation — with global analytic separation constants, a worst-case exponent equal to one isolated-triangle detection, and a consistent NNLS estimator carrying a fully explicit $O(1/N)$ bound. The three regimes delimit both algorithmic structure learning [1, 10, 11] and subspace detection [13]; on real data the same statistic localizes unplanted cyclone circulation. Open problems: identifiability *between* the diagonal and free-PSD excitation classes (e.g. banded or Toeplitz Γ_S — where between reading off every weight and seeing only the image does the transition occur?); sharp first-order sparse thresholds over $\ker \mathbf{B}_2$; effective sample sizes N_{eff} under temporal dependence; and tightening the Markov factor in (5) to match the empirical transition.

7. REFERENCES

- [1] Sravanthi Gurugubelli and Sundeep Prabhakar Chepuri, “Simplicial complex learning from edge flows via sparse clique sampling,” in *European Signal Processing Conference (EUSIPCO)*, 2024.
- [2] Sergio Barbarossa and Stefania Sardellitti, “Topological signal processing over simplicial complexes,” *IEEE Transactions on Signal Processing*, vol. 68, pp. 2992–3007, 2020.
- [3] Michael T. Schaub, Yu Zhu, Jean-Baptiste Seby, T. Mitchell Roddenberry, and Santiago Segarra, “Signal processing on higher-order networks: Livin’ on the edge... and beyond,” *Signal Processing*, vol. 187, pp. 108149, 2021.
- [4] Lek-Heng Lim, “Hodge Laplacians on graphs,” *SIAM Review*, vol. 62, no. 3, pp. 685–715, 2020.
- [5] Maosheng Yang, Elvin Isufi, Michael T. Schaub, and Geert Leus, “Simplicial convolutional filters,” *IEEE Transactions on Signal Processing*, vol. 70, pp. 4633–4648, 2022.
- [6] Claudio Battiloro, Paolo Di Lorenzo, and Sergio Barbarossa, “Topological Slepian: Maximally localized representations of signals over simplicial complexes,” in *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2023.
- [7] Lucille Calmon, Michael T. Schaub, and Ginestra Bianconi, “Dirac signal processing of higher-order topological signals,” *New Journal of Physics*, vol. 25, pp. 093013, 2023.
- [8] Lorenzo Marinucci, Claudio Battiloro, and Paolo Di Lorenzo, “Topological adaptive least mean squares algorithms over simplicial complexes,” *IEEE Transactions on Signal and Information Processing over Networks*, vol. 12, pp. 283–298, 2026, arXiv:2505.23160.
- [9] Chengen Liu, Geert Leus, and Elvin Isufi, “Unrolling of simplicial Elasticnet for edge flow signal reconstruction,” *IEEE Open Journal of Signal Processing*, vol. 5, 2024, DOI 10.1109/OJSP.2023.3339376.
- [10] Andrei Buciulea, Elvin Isufi, Geert Leus, and Antonio G. Marques, “Learning the topology of a simplicial complex using simplicial signals: A greedy approach,” arXiv:2502.20159, 2025.
- [11] Josef Hoppe and Michael T. Schaub, “Representing edge flows on graphs via sparse cell complexes,” arXiv:2309.01632; Learning on Graphs Conference (LoG), 2023.
- [12] Lorenzo Marinucci, Gabriele D’Acunto, Paolo Di Lorenzo, and Sergio Barbarossa, “Simplicial Gaussian models: Representation and inference,” arXiv:2510.12983, 2025.
- [13] Chengen Liu, Victor M. Tenorio, Antonio G. Marques, and Elvin Isufi, “Matched topological subspace detector,” arXiv:2504.05892, 2025.
- [14] H. Vincent Poor, *An Introduction to Signal Detection and Estimation*, Springer, 2nd edition, 1994.
- [15] Thomas M. Cover and Joy A. Thomas, *Elements of Information Theory*, Wiley, 2nd edition, 2006.
- [16] Martin J. Wainwright, “Information-theoretic limits on sparsity recovery in the high-dimensional and noisy setting,” *IEEE Transactions on Information Theory*, vol. 55, no. 12, pp. 5728–5741, 2009.
- [17] Arash A. Amini and Martin J. Wainwright, “High-dimensional analysis of semidefinite relaxations for sparse principal components,” *The Annals of Statistics*, vol. 37, no. 5B, pp. 2877–2921, 2009.
- [18] Quentin Berthet and Philippe Rigollet, “Optimal detection of sparse principal components in high dimension,” *The Annals of Statistics*, vol. 41, no. 4, pp. 1780–1815, 2013.
- [19] T. Tony Cai, Cun-Hui Zhang, and Harrison H. Zhou, “Optimal rates of convergence for covariance matrix estimation,” *The Annals of Statistics*, vol. 38, no. 4, pp. 2118–2144, 2010.
- [20] Xiaoye Jiang, Lek-Heng Lim, Yuan Yao, and Yinyu Ye, “Statistical ranking and combinatorial Hodge theory,” *Mathematical Programming*, vol. 127, no. 1, pp. 203–244, 2011.
- [21] Steven M. Kay, *Fundamentals of Statistical Signal Processing: Estimation Theory*, Prentice-Hall, 1993.
- [22] Hans Hersbach, Bill Bell, Paul Berrisford, et al., “The ERA5 global reanalysis,” *Quarterly Journal of the Royal Meteorological Society*, vol. 146, no. 730, pp. 1999–2049, 2020.
- [23] Kenneth R. Knapp, Michael C. Kruk, David H. Levinson, Howard J. Diamond, and Charles J. Neumann, “The International Best Track Archive for Climate Stewardship (IBTrACS): Unifying tropical cyclone data,” *Bulletin of the American Meteorological Society*, vol. 91, no. 3, pp. 363–376, 2010.

Compliance with ethical standards

Funding and conflicts of interest. This work received no external funding. The author declares no conflicts of interest.

Research involving human participants and/or animals. This article does not contain any studies with human participants or animals. All data used are public: ERA5 reanalysis (Copernicus/ECMWF, via the ARCO-ERA5 mirror), IBTrACS v04r01 (NOAA NCEI), TNTP road networks, and ECB reference exchange rates.

Use of AI tools (disclosure per IEEE/SPS policy). This preprint is an AI-assisted *draft*: substantial portions of the manuscript text, the source code, and the figures were generated with the assistance of an AI system (Claude, Anthropic) under the author’s direction across iterative adversarial-review cycles. Because IEEE/SPS policy does not permit AI-generated manuscript *text*, the author will independently rewrite the manuscript before any formal submission and then disclose only the permitted uses (editing, code, visualization, and review feedback). All mathematical statements are checked by the released test suite (58 tests, including exhaustive and Monte-Carlo checks for every quoted constant), and the full revision history is preserved unaltered in the public repository — git history is deliberately *not* rewritten to obscure AI use. The author takes full responsibility for the content.